Konzentriertheitshypothese

- $d \ll \lambda$ (至少10倍) , $\lambda = \frac{c}{f}$

  $c \approx 3 \times 10^8$ m/s

Resistive Eintore

Beschreibungsform

→ impli. : $f(u,i) = 0$

→ param. : $u = u(\lambda)$, $i = i(\lambda)$

→ expl. (唯一) : { spg. : $i = g(u)$; strg : $u = r(i)$ }

↓

应变量必须命名，才有 expl. Beschr.

Torbedingung

$i_{入} = i_{出}$

Eigenschaften

→ ungepolt : 关于 (0,0) 中心对称

→ quellenfrei : 过 (0,0) [无 ⊖ , ⦶]

→ Leistung $P = ui$ {
- aktiv : $\exists\ ui < 0$
- passiv : $\forall\ ui \geq 0$
- verlustlos : $\forall\ ui = 0$
- verlustbehaftet : $\exists\ ui \neq 0$ }

→ { streng linear : 必过 (0,0) ; (stückweise) linear : 不过 (0,0) }

! aktiv + aktiv 不一定 aktiv, 也可能为 passiv

Dualwandlung

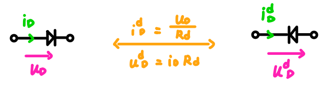
→ $R^d = \frac{R_d^2}{R}$, $G^d = \frac{1}{R_d^2 \cdot G}$

→ Eintor 的 Eigenschaft 无变化

→ strg. ⟷ spg. , 串 ↔ 并 , KS ↔ LL

→ F, F^d 的图像关于 $y = \frac{1}{R_d}$ 对称

→ ideale Diode   ungepolte Diode

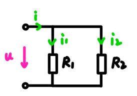


D : { $i_D = 0$, $u_D \leq 0$; $u_D = 0$, $i_D \geq 0$ }

D^d : { $u_D^d = 0$, $i_D^d \leq 0$; $i_D^d = 0$, $u_D^d \geq 0$ }

→ Dualwandlung : $i \to u^d$, $u \to i^d$

Stromteiler



$i_1 = \frac{R_2}{R_1 + R_2} \cdot i = \frac{G_1}{G_1 + G_2} \cdot i$

$i_2 = \frac{R_1}{R_1 + R_2} \cdot i = \frac{G_2}{G_1 + G_2} \cdot i$

streng lineare resistive Eintore

→ Nullator : $u = 0$, $i = 0$

→ Norator : $u = $ bel. , $i = $ bel.

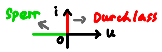
串 : LL
并 : KS

→ Widerstand : $u = Ri$

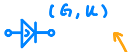
→ Leitwert : $i = Gu$

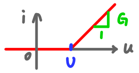
stückweise lineare Widerstand

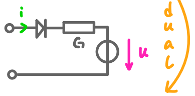
→ ideale Diode : { $i = 0$, $u \leq 0$ **LL** ; $u = 0$, $i \geq 0$ **KS** }



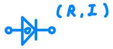
→ konkaver Widerstand :

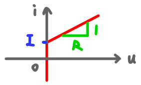


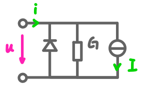




→ konvexer Widerstand :







Anderes

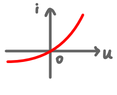
→ LL : $i = 0$, $u = $ bel.

→ KS : $u = 0$, $i = $ bel.

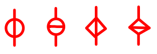
→ Spannungsquelle : $u = U_0$, $i = $ bel.

→ Stromquelle : $i = I_0$, $u = $ bel.

→ pn - Diode : { $i_D = I_S \left(e^{\frac{u_D}{U_T}} - 1\right)$; $u_D = U_T \cdot \ln\left(\frac{i_D}{I_S} + 1\right)$ }



!

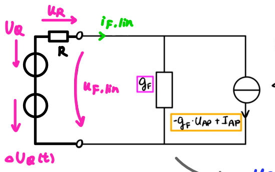


: linear

Linearisierung : G/K ESB

Großsignal ESB

G : $u_{lin} = R(i - I_{AP}) + U_{AP}$



$i_{Flin} = g_F \cdot (u_F - U_{AP}) + I_{AP}$

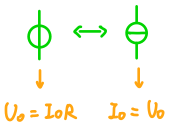
$\Leftrightarrow i_{Flin} = \boxed{g_F \cdot u_F}\ \boxed{- g_F \cdot U_{AP} + I_{AP}}$

(! ähnlich wie konstante Quelle)

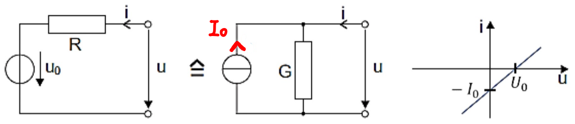
$u_Q(t) = U_Q + \Delta U_Q(t)$

KVL : $u_R + u_g = U_Q + \Delta U_Q(t)$

Quellwandlung linearer Quellen



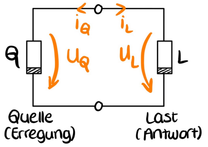
$R \longleftrightarrow G$, 串 ⟷ 并



$u = iR + U_0$ $i = uG - I_0$

U_0 : LL - Spannung I_0 : KS - Strom

Arbeitspunkt bestimmen



KVL : $u_Q = u_L$

KCL : $i_Q = -i_L$

① 画Q图像

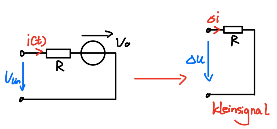
② 将① 按u轴对称得到 Q^x

③ Q^x 与L的交点即为 AP

Linearisierung um Arbeitspunkt

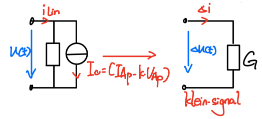
→ stromgesteuert : $u_{F,lin} = \left.\frac{du_F(i_F)}{di_F}\right|_{AP} \underset{\Delta i}{(i_F - I_{AP})} + U_{AP}$

↓ Kleinsignalwiderstand r_F



→ spannungsgesteuert : $i_{F,lin} = \left.\frac{di_F(u_F)}{du_F}\right|_{AP} \underset{\Delta u}{(u_F - U_{AP})} + U_{AP}$

↓ Kleinsignalleitwert g_F



→ Warum ist das Eintor ungepolt ?

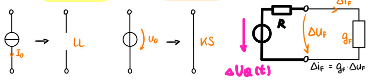
=> Kennlinie ist punktsymmetrisch zum Ursprung .

Kleinsignal ESB

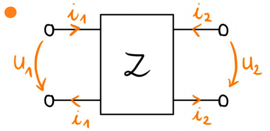
K : $\underbrace{u - U_{AP}}_{\Delta u} = R \underbrace{(i - I_{AP})}_{\Delta i}$

$i_{Flin} = g_F \cdot (u_F - U_{AP}) + I_{AP}$

$\Leftrightarrow \underbrace{(i_{Flin} - I_{AP})}_{\Delta i_F} = g_F \cdot \underbrace{(u_F - U_{AP})}_{\Delta u_F}$ => $\Delta i_F = g_F \cdot \Delta u_F$



Resistive Zweitore



→ Betriebspunkt : $[u_1\ u_2\ i_1\ i_2]^T$

Beschreibungsform

1) implizite Beschreibung:

allgemein: Nullstellenmenge einer Funktion

$$f(\underline{u}, \underline{i}) = \begin{bmatrix} f_1(u_1, u_2, i_1, i_2) \\ f_2(u_1, u_2, i_1, i_2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

nicht eindeutig, existiert aber immer

$f: \mathbb{R}^4 \to \mathbb{R}^2$

streng Linear: Kernbeschreibung (不唯一)

$$F = \text{Kern}[\underset{\sim}{M}\ \ \underset{\sim}{N}] = \left\{ \begin{bmatrix} u \\ i \end{bmatrix} \middle| [\underset{\sim}{M}\ \ \underset{\sim}{N}] \begin{bmatrix} u \\ i \end{bmatrix} = \underline{0} \right\}$$

$\underset{\sim}{M}\underline{u} + \underset{\sim}{N}\underline{i} = 0$ $\underset{\sim}{M}, \underset{\sim}{N} \in \mathbb{R}^{2 \times 2}$

2) parametrische Beschreibung:

allgemein: Beschreibung mithilfe eines Parameterpaares $c_1, c_2 : \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$

$$F = \left\{ \begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} \middle| \exists \underline{c} : \underline{u} = \underline{u}(\underline{c}), \underline{i} = \underline{i}(\underline{c}) \right\}$$

nicht eindeutig, existiert aber immer

streng Linear: Bildbeschreibung

$$F = \left\{ \begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} \middle| \begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} = \begin{bmatrix} \underset{\sim}{U} \\ \underset{\sim}{I} \end{bmatrix} \underline{c} , \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \in \mathbb{R} \right\}$$

若只是 linear, 则要 #Tor +1 个 Messungen

$$\begin{bmatrix} \underset{\sim}{U} \\ \underset{\sim}{I} \end{bmatrix} = \underbrace{\begin{bmatrix} \underline{u}^{(1)} & \underline{u}^{(2)} \\ \underline{i}^{(1)} & \underline{i}^{(2)} \end{bmatrix}}_{\text{Betriebsmatrix}} = \begin{bmatrix} u_1^{(1)} & u_1^{(2)} \\ u_2^{(1)} & u_2^{(2)} \\ i_1^{(1)} & i_1^{(2)} \\ i_2^{(1)} & i_2^{(2)} \end{bmatrix}$$

besteht aus 2 linear unabhängigen Messvektoren

linear unabhängig

$$\begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} = \begin{bmatrix} u^{(1)} - u^{(3)} & u^{(2)} - u^{(3)} \\ i^{(1)} - i^{(3)} & i^{(2)} - i^{(3)} \end{bmatrix} \cdot \underline{c} + \begin{bmatrix} u^{(3)} \\ i^{(3)} \end{bmatrix}$$

→ das Verhalten eines jeden Eintors um einen AP kann sinnvoll durch einen strg. lineares Eintors approximiert werden. (X)

=> eine unabhängige Stromquelle oder ein Elment mit Standardparabel als Kennlinie

im AP (0,0)

explizite Beschreibung:

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \underset{\sim}{R} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} + \underbrace{\begin{bmatrix} u_{1,0} \\ u_{2,0} \end{bmatrix}}_{\text{Quellenwert}}$$

implizite Beschreibung:

$$[\underset{\sim}{M}\ \ \underset{\sim}{N}] \begin{bmatrix} \underline{u} - \underline{u}_0 \\ \underline{i} - \underline{I}_0 \end{bmatrix} = \underline{0}$$

3) explizite Beschreibung:

streng Linear:		allgemein:	
$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \boldsymbol{G} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} g_{11}u_1 + g_{12}u_2 \\ g_{21}u_1 + g_{22}u_2 \end{bmatrix}$	Leitwertsb.	$i_1 = g_1(u_1, u_2)$ $i_2 = g_2(u_1, u_2)$	$\boldsymbol{i} = \boldsymbol{g}(\boldsymbol{u})$
$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \boldsymbol{R} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} r_{11}i_1 + r_{12}i_2 \\ r_{21}i_1 + r_{22}i_2 \end{bmatrix}$	Widerstandsb.	$u_1 = r_1(i_1, i_2)$ $u_2 = r_2(i_1, i_2)$	$\boldsymbol{u} = \boldsymbol{r}(\boldsymbol{i})$
$\begin{bmatrix} u_1 \\ i_2 \end{bmatrix} = \boldsymbol{H} \begin{bmatrix} i_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} h_{11}i_1 + h_{12}u_2 \\ h_{21}i_1 + h_{22}u_2 \end{bmatrix}$	Hybridbeschr.	$u_1 = h_1(i_1, u_2)$ $i_2 = h_2(i_1, u_2)$	$\begin{bmatrix} u_1 \\ i_2 \end{bmatrix} = \boldsymbol{h}\left(\begin{bmatrix} i_1 \\ u_2 \end{bmatrix}\right)$
$\begin{bmatrix} i_1 \\ u_2 \end{bmatrix} = \boldsymbol{H'} \begin{bmatrix} u_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} h'_{11}u_1 + h'_{12}i_2 \\ h'_{21}u_1 + h'_{22}i_2 \end{bmatrix}$	inv. Hybridb.	$i_1 = h'_1(u_1, i_2)$ $u_2 = h'_2(u_1, i_2)$	$\begin{bmatrix} i_1 \\ u_2 \end{bmatrix} = \boldsymbol{h'}\left(\begin{bmatrix} u_1 \\ i_2 \end{bmatrix}\right)$
$\begin{bmatrix} u_1 \\ i_1 \end{bmatrix} = \boldsymbol{A} \begin{bmatrix} u_2 \\ -i_2 \end{bmatrix} = \begin{bmatrix} a_{11}u_2 - a_{12}i_2 \\ a_{21}u_2 - a_{22}i_2 \end{bmatrix}$	Kettenbeschr.	$u_1 = a_1(u_2, -i_2)$ $i_1 = a_2(u_2, -i_2)$	$\begin{bmatrix} u_1 \\ i_1 \end{bmatrix} = \boldsymbol{a}\left(\begin{bmatrix} u_2 \\ -i_2 \end{bmatrix}\right)$
$\begin{bmatrix} u_2 \\ i_2 \end{bmatrix} = \boldsymbol{A'} \begin{bmatrix} u_1 \\ -i_1 \end{bmatrix} = \begin{bmatrix} a'_{11}u_1 - a'_{12}i_1 \\ a'_{21}u_1 - a'_{22}i_1 \end{bmatrix}$	inv. Kettenb.	$u_2 = a'_1(u_1, -i_1)$ $i_2 = a'_2(u_1, -i_1)$	$\begin{bmatrix} u_2 \\ i_2 \end{bmatrix} = \boldsymbol{a'}\left(\begin{bmatrix} u_1 \\ -i_1 \end{bmatrix}\right)$

Eigenschaften

→ aktiv : $\exists\ u^T i < 0 \iff \underset{\sim}{U}^T \underset{\sim}{I} + \underset{\sim}{I}^T \underset{\sim}{U} < 0$

→ passiv : $\forall\ u^T i \geq 0 \iff \underset{\sim}{U}^T \underset{\sim}{I} + \underset{\sim}{I}^T \underset{\sim}{U} \geq 0$

! 矩阵中的所有元素都要 ≥ 0

! 另一个依据： $R > 0$, $G > 0$ $\left(\because P = I^2 R = \frac{U^2}{R}\right)$

! 设 $U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $I = \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$, 则: $U^T I = [u_1\ u_2] \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = u_1 i_1 + u_2 i_2$

$$I^T U = [i_1\ i_2] \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = i_1 u_1 + i_2 u_2$$

→ **verlustlos** : $\forall\ u^T i = 0 \iff \underset{\sim}{U}^T \underset{\sim}{I} + \underset{\sim}{I}^T \underset{\sim}{U} = 0$

$\underset{\sim}{R} = -\underset{\sim}{R}^T$, $\underset{\sim}{G} = -\underset{\sim}{G}^T$

→ **umkehrbar** : $\underset{\sim}{G} = \underset{\sim}{P}\,\underset{\sim}{G}\,\underset{\sim}{P}$, $\underset{\sim}{R} = \underset{\sim}{P}\,\underset{\sim}{R}\,\underset{\sim}{P}$

(symmetrisch)

mit $\underset{\sim}{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$\underset{\sim}{A} = \underset{\sim}{A}^T$

→ **reziprok** : $\underset{\sim}{G} = \underset{\sim}{G}^T$, $\underset{\sim}{R} = \underset{\sim}{R}^T$

$h'_{21} = -h'_{12}$, $h_{21} = -h_{12}$

$r_{21} = r_{12}$, $g_{21} = g_{12}$

$\det \underset{\sim}{A} = \det \underset{\sim}{A}^T = 1$

$\underset{\sim}{U}^T \underset{\sim}{I} - \underset{\sim}{I}^T \underset{\sim}{U} = 0$

Analyse von nicht quellenfreier linearer Zweitor

→ ⦵ → LL , ϕ → KS

→ Zweitormatrix bestimmen : KS-LL Methode

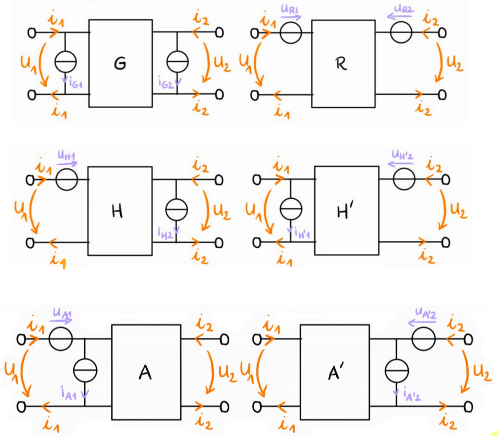
→ Quellenvektor bestimmen : 两个 steuernde Gr. 同时为 0

(KS . LL . Nullator)

此时 gesteuerte Größe 即为 Quellenwert，而后用 KVL. KCL 求出 gesteuerte Gr. ，即 Quellenwert

→ Schaltbild mit den externen Quellen ergänzen :

– zerlegung nicht quellenfreier linearer zweitore:



用来求有源电路中的 expli. Beschr. : $\begin{bmatrix} * \\ * \end{bmatrix} = \begin{bmatrix} * & * \\ * & * \end{bmatrix} \cdot \begin{bmatrix} * \\ * \end{bmatrix} + \boxed{\begin{bmatrix} * \\ * \end{bmatrix}}$

Quellenwert !

Implizit → explizit

$$[\boldsymbol{M}, \boldsymbol{N}] \begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} = \underline{0} \mid \cdot \boldsymbol{M}^{-1}, \qquad [\boldsymbol{M}, \boldsymbol{N}] \begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} = \underline{0} \mid \cdot \boldsymbol{N}^{-1}$$

$$\underline{u} = \underbrace{-\boldsymbol{M}^{-1}\boldsymbol{N}}_{\boldsymbol{R}}\,\underline{i}, \qquad \underline{i} = \underbrace{-\boldsymbol{N}^{-1}\boldsymbol{M}}_{\boldsymbol{G}}\,\underline{u}$$

Explizit → implizit

$$\underline{u} = \boldsymbol{R} \cdot \underline{i}, \qquad \underline{i} = \boldsymbol{G} \cdot \underline{u}$$

$$\underbrace{\boldsymbol{1}}_{\boldsymbol{M}} \cdot \underline{u} \underbrace{-\boldsymbol{R}}_{\boldsymbol{N}} \cdot \underline{i} = \underline{0}, \qquad \underbrace{-\boldsymbol{G}}_{\boldsymbol{M}} \cdot \underline{u} + \underbrace{\boldsymbol{1}}_{\boldsymbol{N}} \cdot \underline{i} = \underline{0}$$

Parametrisiert → explizit

$$\begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} = \begin{bmatrix} \boldsymbol{U} \\ \boldsymbol{I} \end{bmatrix} \underline{c} \Rightarrow \begin{matrix} \underline{u} = \boldsymbol{U}\underline{c} \\ \underline{i} = \boldsymbol{I}\underline{c} \end{matrix}$$

$$\underline{u} = \boldsymbol{U}\underline{c} \mid \cdot \boldsymbol{U}^{-1}, \qquad \underline{i} = \boldsymbol{I}\underline{c} \mid \cdot \boldsymbol{I}^{-1}$$

$$\underline{c} = \boldsymbol{U}^{-1} \cdot \underline{u}, \qquad \underline{c} = \boldsymbol{I}^{-1} \cdot \underline{i}$$

$$\Rightarrow \underline{i} = \underbrace{\boldsymbol{I}\boldsymbol{U}^{-1}}_{\boldsymbol{G}} \cdot \underline{u}, \qquad \Rightarrow \underline{u} = \underbrace{\boldsymbol{U}\boldsymbol{I}^{-1}}_{\boldsymbol{R}} \cdot \underline{i}$$

KS - LL Methode

→ 指出所求的 expli. Beschr.

→ 依次让两个 steuernde Gr. 为 0

→ 用不为 0 的 steuernde Gr. 表示两个 gesteuerte Gr.

Explizit → parametrisiert

$$\underline{u} = \boldsymbol{R} \cdot \underline{i}, \qquad \underline{i} = \boldsymbol{G} \cdot \underline{u}$$

$$\boldsymbol{U} = \boldsymbol{R}; \boldsymbol{I} = \boldsymbol{1}, \qquad \boldsymbol{U} = \boldsymbol{1}; \boldsymbol{I} = \boldsymbol{G}$$

Implizit → parametrisiert

$$\boldsymbol{U} = -\boldsymbol{M}^{-1}\boldsymbol{N}; \boldsymbol{I} = \boldsymbol{1} \text{ oder } \boldsymbol{U} = \boldsymbol{1}; \boldsymbol{I} = -\boldsymbol{N}^{-1}\boldsymbol{M}$$

Parametrisiert → implizit

$$\boldsymbol{M} = -\boldsymbol{I}\boldsymbol{U}^{-1}; \boldsymbol{N} = \boldsymbol{1} \text{ oder } \boldsymbol{M} = \boldsymbol{1}; \boldsymbol{N} = -\boldsymbol{U}\boldsymbol{I}^{-1}$$

Linearisierung von Zweitoren

→ 确认 steuernde . gesteuerte Gr.

→ Jacobimatrix ($\underset{\sim}{J}$) :

$$\begin{bmatrix} u_{1,lin} \\ u_{2,lin} \end{bmatrix} = \underset{\sim}{J}\Big|_{AP} \cdot \begin{bmatrix} i_1 - I_{1,AP} \\ i_2 - I_{2,AP} \end{bmatrix} + \begin{bmatrix} u_{1,AP} \\ u_{2,AP} \end{bmatrix}$$

↓

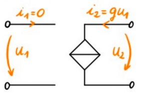
分别对 steuernde Gr. 求导，若式子中依旧存在 steuernde Gr. ，则将 AP 代入

→ Kleinsignal : $\begin{bmatrix} \Delta u_1 \\ \Delta u_2 \end{bmatrix} = \underset{\sim}{J}\Big|_{AP} \cdot \begin{bmatrix} \Delta i_1 \\ \Delta i_2 \end{bmatrix}$

spezielle Zweitore

① gesteuerte Quellen : streng linear !

VCCS (voltage controlled current source)

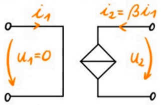


tatsächlich spannungsgesteuert

$\underset{\sim}{G} = \begin{bmatrix} 0 & 0 \\ g & 0 \end{bmatrix}$ $\underset{\sim}{A} = \begin{bmatrix} 0 & -\frac{1}{g} \\ 0 & 0 \end{bmatrix}$ $\mathrm{M} = \begin{bmatrix} 0 & 0 \\ -g & 0 \end{bmatrix}$ $\mathrm{N} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

↳ es gibt nur diese zwei, mehr kann man nicht finden.

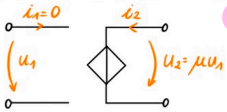
CCCS (current controlled current source)



weder strom- noch spannungsgesteuert

$\underset{\sim}{H} = \begin{bmatrix} 0 & 0 \\ \beta & 0 \end{bmatrix}$ $\underset{\sim}{A} = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{\beta} \end{bmatrix}$

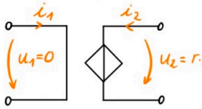
VCVS (voltage controlled voltage source)



weder strom- noch spannungsgesteuert

$\underset{\sim}{H}' = \begin{bmatrix} 0 & 0 \\ \mu & 0 \end{bmatrix}$ $\underset{\sim}{A} = \begin{bmatrix} \frac{1}{\mu} & 0 \\ 0 & 0 \end{bmatrix}$

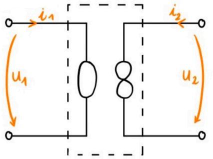
CCVS (current controlled voltage source)



tatsächlich stromgesteuert

$\underset{\sim}{R} = \begin{bmatrix} 0 & 0 \\ r & 0 \end{bmatrix}$ $\underset{\sim}{A} = \begin{bmatrix} 0 & 0 \\ \frac{1}{r} & 0 \end{bmatrix}$ $\mathrm{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\mathrm{N} = \begin{bmatrix} 0 & 0 \\ -r & 0 \end{bmatrix}$

② Nullor : quellenfrei . streng linear . aktiv !

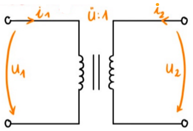


$u_1 = 0\,\mathrm{V}$ u_2 beliebig

$i_1 = 0\,\mathrm{A}$ i_2 beliebig

$\underset{\sim}{A}_{Nullor} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $\mathrm{M} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $\mathrm{N} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

③ Übertrager : verlustlos , reziprok , 无 $\underset{\sim}{G}$. $\underset{\sim}{R}$!



ideal : $ü = \pm 1$

$u_1 = ü \cdot u_2$ $i_1 = \frac{1}{ü}(-i_2)$

$\mathcal{F}_ü = \left\{(u_1, i_1) \mid (u_2, -i_2) = \left(\frac{u_1}{ü}, ü\, i_1\right) \in \mathcal{F}\right\}$

$\underset{\sim}{A} = \begin{bmatrix} ü & 0 \\ 0 & \frac{1}{ü} \end{bmatrix}$ $\underset{\sim}{H} = \begin{bmatrix} 0 & ü \\ -ü & 0 \end{bmatrix}$

$\underset{\sim}{A}' = \begin{bmatrix} \frac{1}{ü} & 0 \\ 0 & ü \end{bmatrix}$ $\underset{\sim}{H}' = \begin{bmatrix} 0 & -\frac{1}{ü} \\ \frac{1}{ü} & 0 \end{bmatrix}$

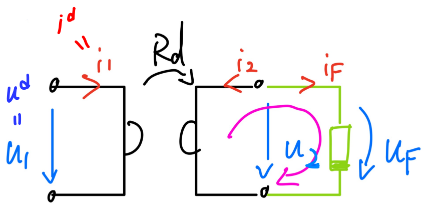
Ohne $\underset{\sim}{R}$ und $\underset{\sim}{G}$ (falls existieren, sind Null) :

falls $\underset{\sim}{R}$ ($\underset{\sim}{R} = \underset{\sim}{0}$) ⇒ 2 KS

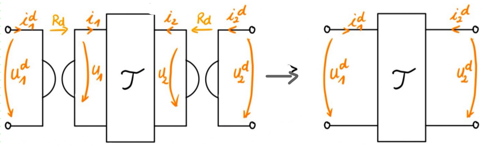
falls $\underset{\sim}{G}$ ($\underset{\sim}{G} = \underset{\sim}{0}$) ⇒ 2 LL

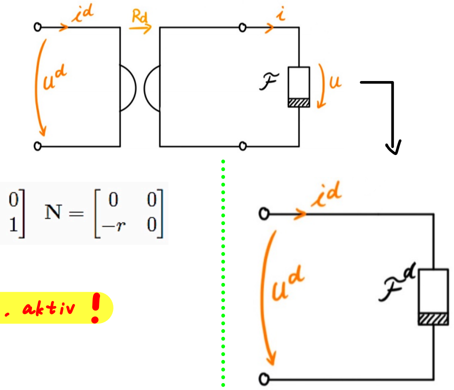
④ Gyrator (Dualwandler) : verlustlos , nicht reziprok !

无 $\underset{\sim}{H}$. $\underset{\sim}{H}'$!



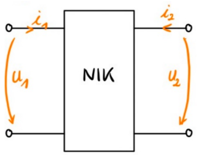
$$\begin{cases} u_2 = R_d\, i_F = R_d\, i_1 \\ i_F = -i_2 = \frac{u_1}{R_d} \end{cases}$$





⑤ Negativ - Immitanz - Konverter (NIK) :

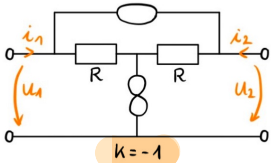
weder strg. noch spg., 无 $\underset{\sim}{R}$. $\underset{\sim}{G}$!



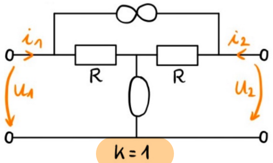
$u_1 = -k u_2$, $i_1 = -\frac{1}{k} i_2$

$\underset{\sim}{H} = \begin{bmatrix} 0 & -k \\ k & 0 \end{bmatrix}$ $\quad$ $\underset{\sim}{A} = \begin{bmatrix} -k & 0 \\ 0 & \frac{1}{k} \end{bmatrix}$

$\underset{\sim}{H}' = \begin{bmatrix} 0 & -\frac{1}{k} \\ \frac{1}{k} & 0 \end{bmatrix}$ $\quad$ $\underset{\sim}{A}' = \begin{bmatrix} -\frac{1}{k} & 0 \\ 0 & k \end{bmatrix}$



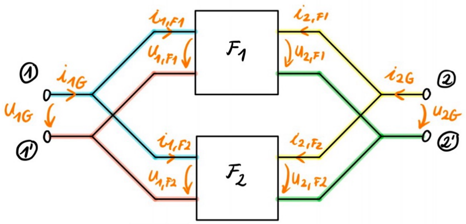
k = -1



k = 1

Zusammenschaltung von Zweitoren

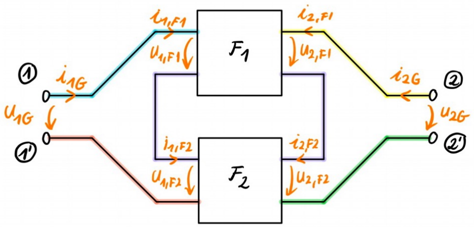
Parallelschaltung



$g_G = g_{F1} + g_{F2}$

streng linear : $\underset{\sim}{G}_G = \underset{\sim}{G}_{F1} + \underset{\sim}{G}_{F2}$

Serienschaltung

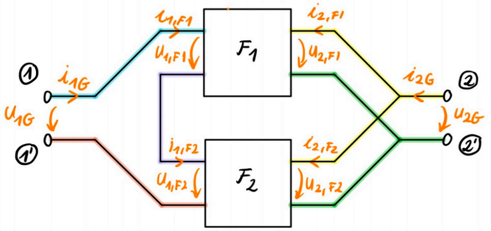


$r_G = r_{F1} + r_{F2}$

streng linear : $\underset{\sim}{R}_G = \underset{\sim}{R}_{F1} + \underset{\sim}{R}_{F2}$

Hybride Verschaltungen

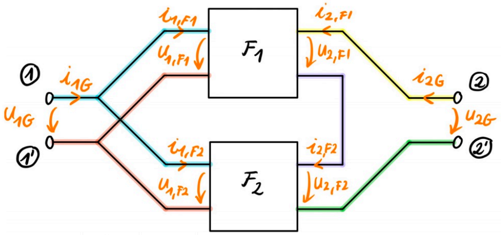
Serien- Parallel- Schaltung



$h_G = h_{F1} + h_{F2}$

streng linear : $\underset{\sim}{H}_G = \underset{\sim}{H}_{F1} + \underset{\sim}{H}_{F2}$

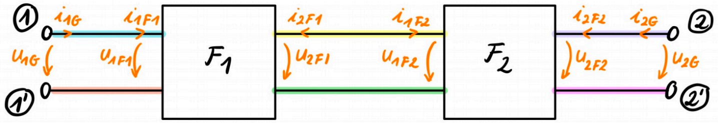
Parallel- Serien- Schaltung



$h'_G = h'_{F1} + h'_{F2}$

streng linear : $\underset{\sim}{H}'_G = \underset{\sim}{H}'_{F1} + \underset{\sim}{H}'_{F2}$

Kettenschaltung



$a_G = a_{\underline{F1}} \circ a_{\underline{F2}}$

streng linear :

$\underset{\sim}{A}_G = \underset{\sim}{A}_{F1} \ \underset{\sim}{A}_{F2}$

beachte die Reihenfolge (da nicht kommutativ!)

$a'_G = a'_{\underline{F2}} \circ a'_{\underline{F1}}$

streng linear :

$\underset{\sim}{A}'_G = \underset{\sim}{A}'_{F2} \ \underset{\sim}{A}'_{F1}$

inverse Kettendarstellung

$\underline{a}'_{ges.} = \underline{a}'_i \circ \underline{a}'_i$

streng linear · $\underset{\sim}{A}'_G = \underset{\sim}{A}'_{F1} \cdot \underset{\sim}{A}'_{F2}$

• Kurzfragen

→ $\boxed{z.B}$ streng lineare Zweitor mit $\underset{\sim}{R} = \begin{bmatrix} 4\Omega & 3\Omega \\ 3\Omega & 5\Omega \end{bmatrix}$

=> nicht verlustlos : $\underset{\sim}{I} = 1A \cdot \underset{\sim}{1}$ => $\underset{\sim}{U} = 1A \cdot \underset{\sim}{R}$

$\therefore \underset{\sim}{U}^T \underset{\sim}{I} + \underset{\sim}{I}^T \underset{\sim}{U} = 1A \cdot \underset{\sim}{R}^T \cdot 1A \cdot \underset{\sim}{1} + 1A \cdot \underset{\sim}{1} \cdot 1A \cdot \underset{\sim}{R}$

$= 1A^2 \cdot \underset{\sim}{R}^T + 1A^2 \ \underset{\sim}{R} = \begin{bmatrix} 8VA & 6VA \\ 6VA & 10VA \end{bmatrix} \neq \underset{\sim}{0}$

Betriebsmatrix

→ spg. : $\underset{\sim}{U}^{-1}$ 存在

det $\underset{\sim}{R} \neq 0$

→ strg. : $\underset{\sim}{I}^{-1}$ 存在

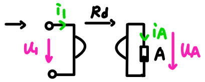
det $\underset{\sim}{G} \neq 0$

→ Die Linearisierung eines Gyrators ist der selbe Gyrator.

→ VCCS $\underline{\text{dual}}$ CCVS

→ Die Übertragungscharakteristik einer Schaltung kann um einen beliebigen AP linearisiert werden. (X)

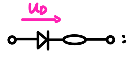
=> 还必须 bestimmen, im AP 要 stetig und differenzierbar

→ 

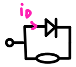
Was ist an Tor 1 :

das duale Eintor zu A

$u_1 = u^d_A$, $i_1 = i^d_A$

→ 

: LL mit $u_F < 0$



: KS mit $i_D > 0$ (Diode im Durlassber.)

→ KS - LL Methode 能用的原因 : Die Messungen lin. unab.

(Messungen sinnvoll , um param. Beschr. von Zweitor zu finden)

→ Zusammenschaltung von Zweitor ist möglich 的前提条件 :

Die Torbedingungen an allen 6 Toren (视乎问题而异)

→ 求 Innenwiderstand Ri : ① Quellen = 0

(Gi 同理) ② 列 KCL . KVL 得出 $\frac{u_F}{i_F}$, 即为 Ri

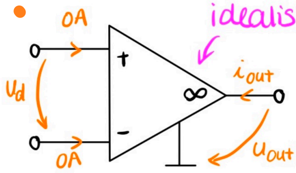
→ quellenfrei 的原因 : Schaltung hat nur streng lineare Elemente

gegeben

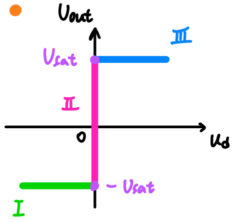
	$\boldsymbol{R}$	$\boldsymbol{G}$	$\boldsymbol{H}$	$\boldsymbol{H'}$	$\boldsymbol{A}$	$\boldsymbol{A'}$
$\boldsymbol{R}$	$\begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}$	$\frac{1}{\det \boldsymbol{G}}\begin{bmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{bmatrix}$	$\frac{1}{h_{22}}\begin{bmatrix} \det \boldsymbol{H} & h_{12} \\ -h_{21} & 1 \end{bmatrix}$	$\frac{1}{h'_{11}}\begin{bmatrix} 1 & -h'_{12} \\ h'_{21} & \det \boldsymbol{H'} \end{bmatrix}$	$\frac{1}{a_{21}}\begin{bmatrix} a_{11} & \det \boldsymbol{A} \\ 1 & a_{22} \end{bmatrix}$	$\frac{1}{a'_{21}}\begin{bmatrix} a'_{22} & 1 \\ \det \boldsymbol{A'} & a'_{11} \end{bmatrix}$
$\boldsymbol{G}$	$\frac{1}{\det \boldsymbol{R}}\begin{bmatrix} r_{22} & -r_{12} \\ -r_{21} & r_{11} \end{bmatrix}$	$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$	$\frac{1}{h_{11}}\begin{bmatrix} 1 & -h_{12} \\ h_{21} & \det \boldsymbol{H} \end{bmatrix}$	$\frac{1}{h'_{22}}\begin{bmatrix} \det \boldsymbol{H'} & h'_{12} \\ -h'_{21} & 1 \end{bmatrix}$	$\frac{1}{a_{12}}\begin{bmatrix} a_{22} & -\det \boldsymbol{A} \\ -1 & a_{11} \end{bmatrix}$	$\frac{1}{a'_{12}}\begin{bmatrix} a'_{11} & -1 \\ -\det \boldsymbol{A'} & a'_{22} \end{bmatrix}$
$\boldsymbol{H}$	$\frac{1}{r_{22}}\begin{bmatrix} \det \boldsymbol{R} & r_{12} \\ -r_{21} & 1 \end{bmatrix}$	$\frac{1}{g_{11}}\begin{bmatrix} 1 & -g_{12} \\ g_{21} & \det \boldsymbol{G} \end{bmatrix}$	$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$	$\frac{1}{\det \boldsymbol{H'}}\begin{bmatrix} h'_{22} & -h'_{12} \\ -h'_{21} & h'_{11} \end{bmatrix}$	$\frac{1}{a_{22}}\begin{bmatrix} a_{12} & \det \boldsymbol{A} \\ -1 & a_{21} \end{bmatrix}$	$\frac{1}{a'_{11}}\begin{bmatrix} a'_{12} & 1 \\ -\det \boldsymbol{A'} & a'_{21} \end{bmatrix}$
$\boldsymbol{H'}$	$\frac{1}{r_{11}}\begin{bmatrix} 1 & -r_{12} \\ r_{21} & \det \boldsymbol{R} \end{bmatrix}$	$\frac{1}{g_{22}}\begin{bmatrix} \det \boldsymbol{G} & g_{12} \\ -g_{21} & 1 \end{bmatrix}$	$\frac{1}{\det \boldsymbol{H}}\begin{bmatrix} h_{22} & -h_{12} \\ -h_{21} & h_{11} \end{bmatrix}$	$\begin{bmatrix} h'_{11} & h'_{12} \\ h'_{21} & h'_{22} \end{bmatrix}$	$\frac{1}{a_{11}}\begin{bmatrix} a_{21} & -\det \boldsymbol{A} \\ 1 & a_{12} \end{bmatrix}$	$\frac{1}{a'_{22}}\begin{bmatrix} a'_{21} & -1 \\ \det \boldsymbol{A'} & a'_{12} \end{bmatrix}$
$\boldsymbol{A}$	$\frac{1}{r_{21}}\begin{bmatrix} r_{11} & \det \boldsymbol{R} \\ 1 & r_{22} \end{bmatrix}$	$\frac{1}{g_{21}}\begin{bmatrix} -g_{22} & -1 \\ -\det \boldsymbol{G} & -g_{11} \end{bmatrix}$	$\frac{1}{h_{21}}\begin{bmatrix} -\det \boldsymbol{H} & -h_{11} \\ -h_{22} & -1 \end{bmatrix}$	$\frac{1}{h'_{21}}\begin{bmatrix} 1 & h'_{22} \\ h'_{11} & \det \boldsymbol{H'} \end{bmatrix}$	$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$	$\frac{1}{\det \boldsymbol{A'}}\begin{bmatrix} a'_{22} & a'_{12} \\ a'_{21} & a'_{11} \end{bmatrix}$
$\boldsymbol{A'}$	$\frac{1}{r_{12}}\begin{bmatrix} r_{22} & \det \boldsymbol{R} \\ 1 & r_{11} \end{bmatrix}$	$\frac{1}{g_{12}}\begin{bmatrix} -g_{11} & -1 \\ -\det \boldsymbol{G} & -g_{22} \end{bmatrix}$	$\frac{1}{h_{12}}\begin{bmatrix} 1 & h_{11} \\ h_{22} & \det \boldsymbol{H} \end{bmatrix}$	$\frac{1}{h'_{12}}\begin{bmatrix} -\det \boldsymbol{H'} & -h'_{22} \\ -h'_{11} & -1 \end{bmatrix}$	$\frac{1}{\det \boldsymbol{A}}\begin{bmatrix} a_{22} & a_{12} \\ a_{21} & a_{11} \end{bmatrix}$	$\begin{bmatrix} a'_{11} & a'_{12} \\ a'_{21} & a'_{22} \end{bmatrix}$

gesucht

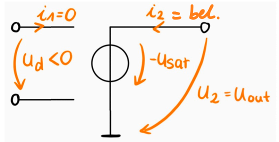
Operationsverstärker


OA
idealisiert
+
∞
−
i_{out}
U_d
OA
U_{out}

- −: invertierender Eingang
- +: nicht invertierender Eingang
- U_d zeigt immer von + nach −
- I_{out} ist beliebig und Eingangsströme sind OA.



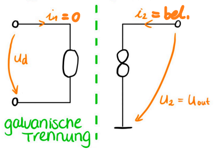
(I): negative Sättigung


$i_1=0$
i_2 = bel.
$U_d<0$
$-U_{sat}$
$U_2=U_{out}$

$U_{out} = -U_{sat}$

$U_d < 0$

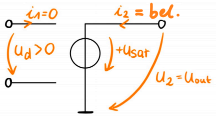
(II): linearer Bereich


$i_1=0$
i_2 = bel.
U_d
$U_2=U_{out}$
galvanische Trennung

$|U_{out}| \leq U_{sat}$

$U_d = 0$

(III): positive Sättigung


$i_1=0$
i_2 = bel.
$U_d>0$
$+U_{sat}$
$U_2=U_{out}$

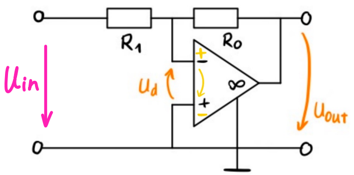
$U_{out} = U_{sat}$

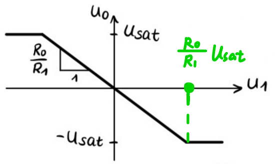
$U_d > 0$

Operationsverstärkerschaltungen

① **invertierender Verstärker**:

$$U_{out} = \begin{cases} U_{sat} \\ -\frac{R_0}{R_1} U_{in} \\ -U_{sat} \end{cases}$$

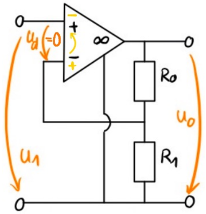

U_{in}
R_1
R_0
U_d
−
+
∞
U_{out}

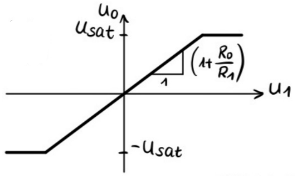


$V = \frac{U_0}{U_1} = \left(-\frac{R_0}{R_1}\right)$ Spannungsverstärkung

② **nicht invertierender Verstärker**:

$$U_{out} = \begin{cases} -U_{sat} \\ \left(1+\frac{R_0}{R_1}\right) U_{in} \\ U_{sat} \end{cases}$$

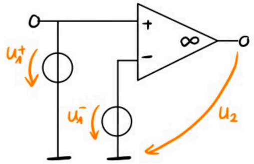

$U_d=0$
+
−
∞
R_0
U_0
R_1
U_1



$V = \frac{U_0}{U_1} = \left(1+\frac{R_0}{R_1}\right)$ Spannungsverstärkung

③ **Komparator**:

$$U_{out} = \begin{cases} U_{sat}, & U_1^+ > U_1^- \\ -U_{sat}, & U_1^+ < U_1^- \end{cases}$$

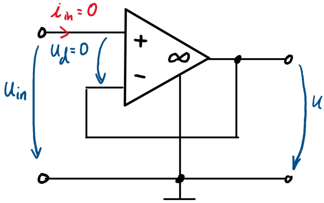


用于: Impedanzwandlung

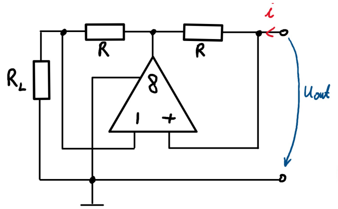
④ **Spannungsfolger**:

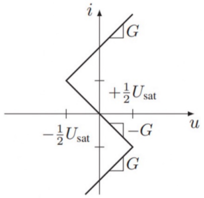
$U_{in} = U_{out}$

$i_{in} = 0$


$i_{in}=0$
$U_d=0$
+
∞
−
U_{in}
U_{out}

⑤ **NIK**:

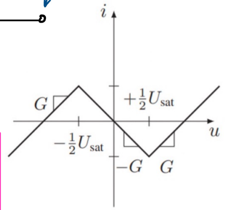




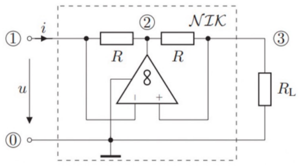
„S“-Typ

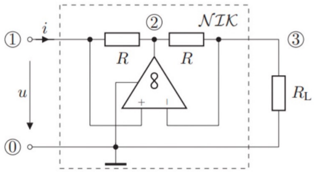
$U_1 = -kU_2$

$i_1 = -\frac{1}{k} i_2$



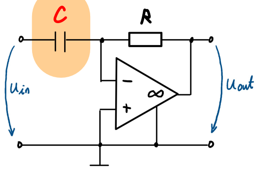
„N“-Typ


①
②
NIK
③
u
R
R
∞
R_L
⓪


①
②
NIK
③
u
R
R
∞
R_L
⓪

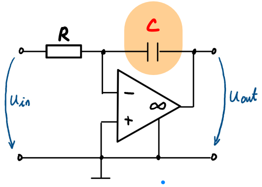
⑥ **Differenzierer**:

$U_{out} = -RC \cdot \dot{U}_{in}$



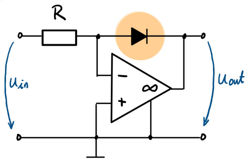
⑦ **Integrierer**:

$U_{out} = \frac{1}{-RC} \int_{t_0}^{t_1} U_{in}\, dt$



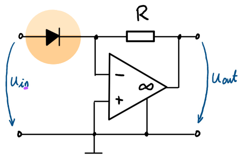
⑧ **Logarithmierer**:

$U_{out} = -U_T \cdot \ln\left(\frac{U_{in}}{R \cdot I_S}\right)$



⑨ **Exponentierer**:

$U_{out} = -RI_S \cdot \exp\left(\frac{U_{in}}{U_T}\right)$



→ $U_{out} = 4(U_1 - U_2)$: Differenzverstärker mit $V = 4$

Resistive Mehrtore

- Bestehen aus p-Tore, die die Torbedingung erfüllen!
- $\frac{(2p)!}{(p!)^2}$ Möglichkeiten für explizite Beschreibung
- Für Kettenbeschreibung gerade Anzahl an Toren
- Kettenverschaltung analog zu Zweitoren

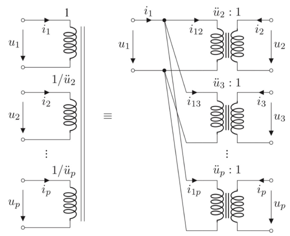
spezielle Mehrtore

① **Mehrtorübertrager**:

$\boxed{H_ü = -H_ü^T}$

verlustlos: $\underset{\sim}{U}^T \underset{\sim}{I} + \underset{\sim}{I}^T \underset{\sim}{U} = \underset{\sim}{0}$!

reziprok: $\underset{\sim}{U}^T \underset{\sim}{I} - \underset{\sim}{I}^T \underset{\sim}{U} = \underset{\sim}{0}$!


1
i_1
u_1
$1/ü_2$
i_2
u_2
$1/ü_p$
i_p
u_p
≡
i_1
$ü_2 : 1$
i_{12}
i_2
u_2
$ü_3 : 1$
i_{13}
i_3
u_3
$ü_p : 1$
i_{1p}
i_p
u_p

$$\begin{bmatrix} i_a \\ u_b \end{bmatrix} = \underset{\sim}{H} \begin{bmatrix} u_a \\ i_b \end{bmatrix}$$

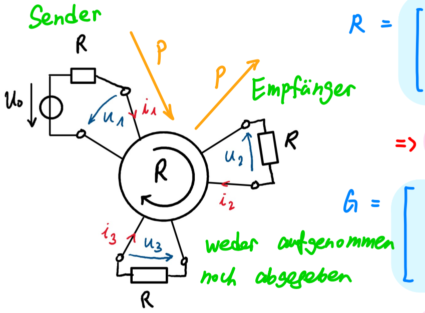
→ u_a, i_a sind die ersten P_1 Tore

$\boxed{zB}$ $\begin{bmatrix} i_1 \\ u_2 \\ \vdots \\ u_p \end{bmatrix} = \underset{\sim}{H} \cdot \begin{bmatrix} u_1 \\ i_2 \\ \vdots \\ i_p \end{bmatrix}$

$$\underset{\sim}{H}_{über.} = \begin{bmatrix} 0 & H_{a,b} \\ -\underset{\sim}{H}^T_{a,b} & 0 \end{bmatrix}$$

$\boxed{z.B}$ $\begin{bmatrix} i_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{ü_2} & \cdots & -\frac{1}{ü_n} \\ \frac{1}{ü_2} & & \underset{\sim}{0} & \\ \vdots & & & \\ \frac{1}{ü_n} & & & \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ i_2 \\ \vdots \\ i_n \end{bmatrix}$

② Zirkulator : verlustlos , nicht reziprok !



weder aufgenommen noch abgegeben

$$\underset{\sim}{R} = \begin{bmatrix} 0 & R & -R \\ -R & 0 & R \\ R & -R & 0 \end{bmatrix} \Rightarrow \underset{\sim}{R} = -\underset{\sim}{R}^T$$

$$\underset{\sim}{G} = \begin{bmatrix} 0 & G & -G \\ -G & 0 & G \\ G & -G & 0 \end{bmatrix} \Rightarrow \underset{\sim}{G} = -\underset{\sim}{G}^T$$

$$\boxed{P_1 = -P_2 \ , \ P_3 = 0}$$

→ an Tor 1 : Leistung aufgenommen werden

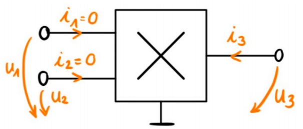
→ an Tor 2 : Leistung wieder abgegeben werden

→ an Tor 3 : Leistung weder aufgenommen noch abgegeben

→ $u_1 = \frac{u_0}{2}$, $u_2 = -\frac{u_0}{2}$, $u_3 = 0$

→ $i_1 = \frac{u_0}{2R}$, $i_2 = \frac{u_0}{2R}$, $i_3 = 0$

③ Multiplizierer :



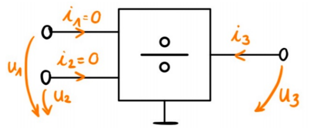
$$\begin{bmatrix} i_1 \\ i_2 \\ u_3 \end{bmatrix} = h\left(\begin{bmatrix} u_1 \\ u_2 \\ i_3 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ \frac{u_1 \cdot u_2}{U_M} \end{bmatrix}$$

Multipliziererkonstante

↓

steuernde Gr. 的组合： $u_1 . u_2 . u_3$ 的其中 2 个 + i_3

④ Dividierer :



$$\begin{bmatrix} i_1 \\ i_2 \\ u_3 \end{bmatrix} = h\left(\begin{bmatrix} u_1 \\ u_2 \\ i_3 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ \frac{u_1}{u_2} U_D \end{bmatrix}$$

Dividiererkonstante

→ 判断 Schaltung 是否 plausibel : (已知 $\underset{\sim}{A}$. $\underset{\sim}{B}$)

$\boxed{法 1}$: $\underset{\sim}{B} \cdot \underset{\sim}{A}^T = A \cdot \underset{\sim}{B}^T = 0$

$\boxed{法 2}$: $\underset{\sim}{B} = -\underset{\sim}{A}^T$

(前提条件 : $\underset{\sim}{A}$ 、$\underset{\sim}{B}$ 是用 Baumkonzept 算出来的)

Allgemeine Analyseverfahren

Baum

$\boxed{不唯一}$

- ist zusammenhängend
- enthält alle Knoten
- verbindet alle Knoten

→ hat (n-1) Zweige

Baumkonzept

KVL
- #: b-(n-1)
- Verbindungskante × 1

→ Nullraumdar. : $\underset{\sim}{B} \cdot \underline{u} = [\underset{\sim}{B}_t \ . \ \underset{\sim}{1}_s] \cdot \begin{bmatrix} u_t \\ u_l \end{bmatrix} = 0$

→ Bildraumdar. : $\underline{u} = \underset{\sim}{A}^T \underline{u}_t$ (Baumzweigspannungen)

KCL
- #: n-1
- Baumkante × 1

→ Nullraumdar. : $\underset{\sim}{A} \cdot \underline{i} = [\underset{\sim}{1}_{n-1} \ . \ -B_t^T] \cdot \begin{bmatrix} i_t \\ i_l \end{bmatrix} = 0$

→ Bildraumdar. : $\underline{i} = \underset{\sim}{B}^T \underline{i}_l$ (Verbindungskantenströme)

! Durch Baumkonzept muss : $\underset{\sim}{A}_l = -\underset{\sim}{B}_t^T$

Knoteninzidenzmatrix $\underset{\sim}{A}$

! $\underset{\sim}{A}$ 的每一列的和均为 0，最后一行为 Bezugsknoten

! Bezugsknoten = Masse

→ $\underset{\sim}{A} \cdot \underline{i} = 0$

→ Zweigspannungen : u ; Knotenspannungen : u_k

则 : $\underline{u} = \underset{\sim}{A}^T \underline{u}_k$

! 已知 Knoteninzidenzmatrix $\underset{\sim}{A}$, 求 zu $\underset{\sim}{A}$ passende Matrix $\underset{\sim}{B}$: $\underset{\sim}{A}\underset{\sim}{B}^T = 0$

Tellengenscher Satz

→ $\underline{u}^T \cdot \underline{i} = \underline{0}$ —— 判断是否为 Betriebspunkt

→ $\underset{\sim}{A}\underset{\sim}{B}^T = \underset{\sim}{B}\underset{\sim}{A}^T = 0$

Tableaugleichungen

$$\begin{array}{r} \text{KVL} \rightarrow \\ \text{KCL} \rightarrow \\ \text{Elementgl.} \rightarrow \end{array} \underbrace{\begin{bmatrix} \boldsymbol{B} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{A} \\ \boldsymbol{M} & \boldsymbol{N} \end{bmatrix}}_{T \in \mathbb{R}^{2bx2b}} \begin{bmatrix} \underline{u} \\ \underline{i} \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{0} \\ \underline{e} \end{bmatrix} \begin{array}{l} s=b-(n-1) \\ n-1 \\ b \end{array} \Bigg\} 2b \text{ Gl.}$$

↓

圆形／菱形 Quelle × 1 , Nullator × 2

→ 它的 duale Netzwerk :

$$\begin{bmatrix} A & 0 \\ 0 & B \\ \frac{N}{Rd} & \frac{M}{Rd} \end{bmatrix} \cdot \begin{bmatrix} \underline{u}^d \\ \underline{i}^d \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{0} \\ \underline{e} \end{bmatrix}$$

Newton - Ralphson - Algorithmus

① nicht lineare Tableau-Gleichugen :

$$\begin{bmatrix} B & 0 \\ 0 & A \\ M' & N' \end{bmatrix} \begin{bmatrix} u \\ i \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ e \end{bmatrix} = e'$$
$$f'(u,i) = 0.$$

② 对 $f'(u,i)$ 求 $\underset{\sim}{J}(x^{(k)})$; 已知 $x^{(0)}$

③ 根据 $x^{(0)}$, 求出 $f'(x^{(0)})$, $\underset{\sim}{J}(x^{(0)})$

④ 列出 $f'(x^{(0)}) + \underset{\sim}{J}(x^{(0)}) \cdot (x^{(1)} - x^{(0)}) = 0$, 得出 $u_1^{(1)}, u_2^{(1)}, i_1^{(1)}, i_1^{(2)}$ 的等式关系，代入 Tableau，从而补全 Tableau，得到 lineare Tableau-Gleichugen

⑤ 根据 lineare Tableau-Gleichugen 的 4 个等式求出 $u_1^{(1)}, u_2^{(1)}, i_1^{(1)}, i_1^{(2)}$，即 $x^{(1)}$

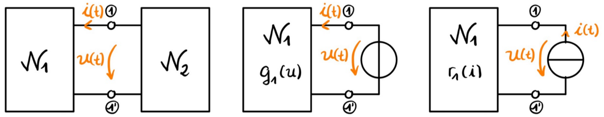
⑥ 同理，求出 $x^{(2)}$, $x^{(3)}$, …

$f(x^{(j)}) + \underset{\sim}{J}(x^{(j)}) \cdot (x^{(j+1)} - x^{(j)}) = 0$

☆ 若为 Eindimension : $x^{(j+1)} = x^{(j)} - \dfrac{f(x^{(j)})}{\frac{d}{dx} f(x^{(j)})}$

Substitutionstheorem

N₁ spannungsgesteuert	⇨	N₂ durch Spannungsquelle ersetzen
N₁ stromgesteuert	⇨	N₂ durch Stromquelle ersetzen



Superpositionsprinzip

→ 使用条件 : Schaltung hat nur linearen Elementen

→ 在 Superpositionsprinzip 中
- passiv : KS $I_0 = 0$
- aktiv : KS $I_0 \neq 0$

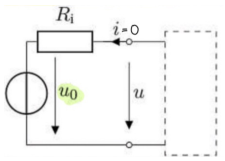
→ Reaktionen der einzelnen Quellen betrachten: jeweils alle Quellen bis auf eine durch ihre Nullquelle ersetzen und gesuchte Größe in Abh. von der Quelle ausdrücken

→ Teillösungen addieren

→ Die Superpositionsprinzip gilt bei der komplexen Wechselstromrechnung und einer Erregung durch zwei Quellen mit unterschiedlichen Kreisfrequenzen für die Zeitsignale.

Zweipolersatzschaltungen

① **Helmholt - Thévenin** :

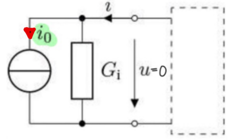


u_0 : LL-Spannungen

R_i : Innenwiderstand

$R_i = -\frac{U_0}{I_0}$

② **Mayor - Norzon** :



I_0 : KS-Strom

G_i : Innenleitwert

$G_i = -\frac{I_0}{U_0}$

→ Wieso kann man statt dem Eintor L das Mayer-Norton ESB verwenden, um AP zu bestimmen?

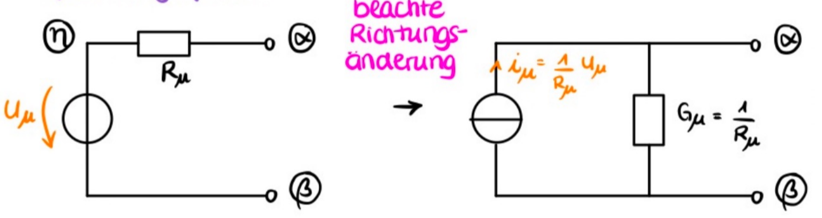
=> Substitutionstheorem und Q ist spannungsgesteuert.

Reduzierte Knotenspannungsanalyse

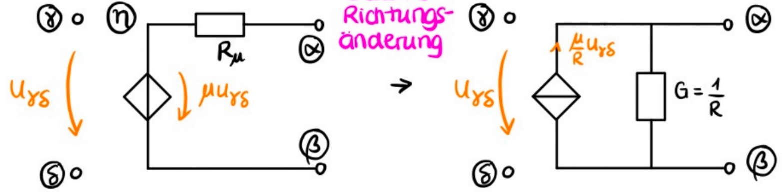
- Umwandlung nicht spg. Elementer

⓪ Quellwandlung :

Spannungsquelle :

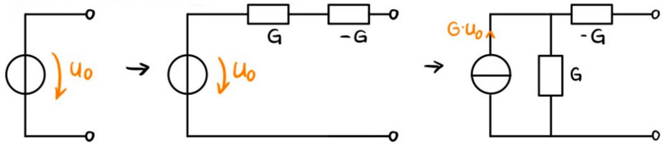

η
R_μ
α
u_μ
β
beachte Richtungsänderung
→
$i_\mu = \frac{1}{R_\mu} u_\mu$
$G_\mu = \frac{1}{R_\mu}$
α
β

VCVS:


γ
η
R_μ
α
$u_{\gamma\delta}$
$\mu u_{\gamma\delta}$
β
δ
beachte Richtungsänderung
→
VCCS
γ
α
$\frac{\mu}{R} u_{\gamma\delta}$
$u_{\gamma\delta}$
$G = \frac{1}{R}$
δ
β

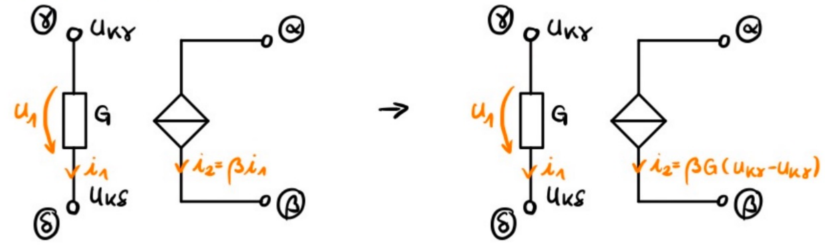
← nur bei voltage controlled

② Quellenwandlung mit Trick :


u_0
→
G
-G
u_0
→
$G \cdot u_0$
-G
G

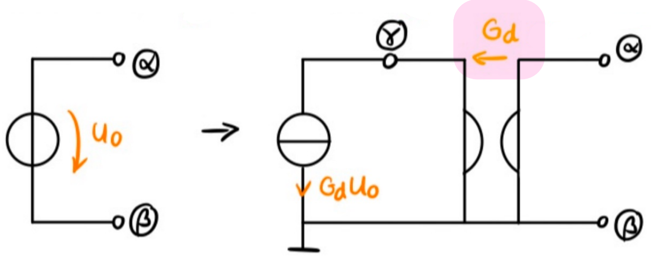
③ Ohmsches Gesetz :

CCCS → VCCS

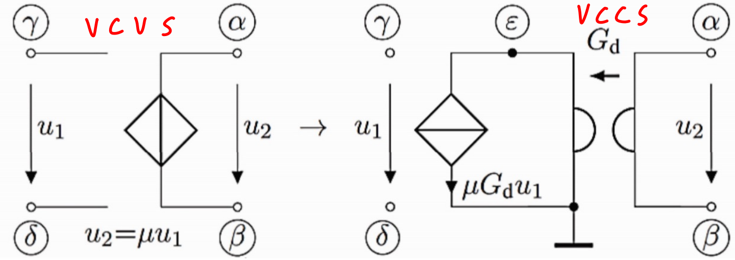

γ
$u_{k\gamma}$
α
u_1
G
i_1
$u_{k\delta}$
δ
$i_2 = \beta i_1$
β
→
γ
$u_{k\gamma}$
α
u_1
G
i_1
$u_{k\delta}$
δ
$i_2 = \beta G (u_{k\gamma} - u_{k\delta})$
β

④ Dualwandlung mit einem Gyrator :

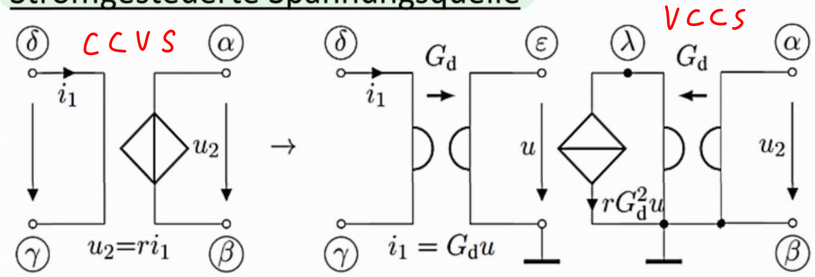
Gyrator :
$$\begin{cases} i_1 = G_d u_2 \\ i_2 = -G_d u_1 \end{cases}$$


α
u_0
β
→
γ
G_d
α
$G_d u_0$
β

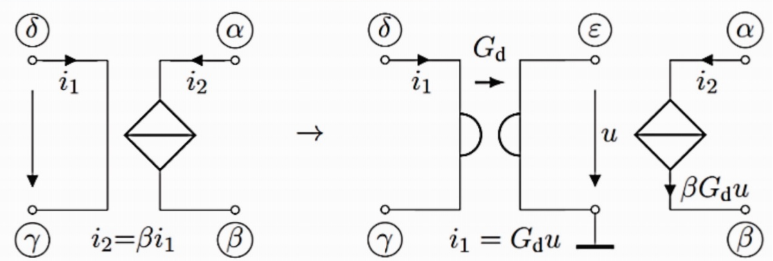
Spannungsgesteuerte Spannungsquelle


VCVS
γ
α
u_1
u_2
δ
$u_2 = \mu u_1$
β
→
VCCS
γ
ε
G_d
α
u_1
$\mu G_d u_1$
u_2
δ
β

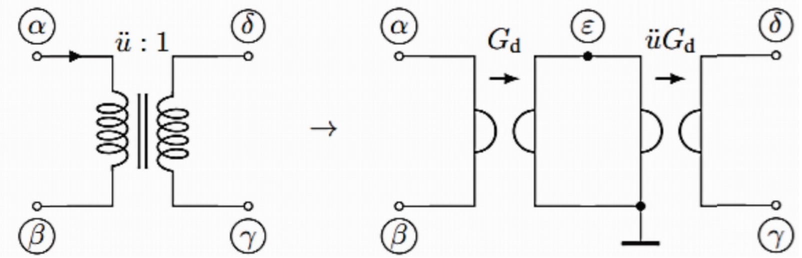
Stromgesteuerte Spannungsquelle


CCVS
δ
α
i_1
u_2
γ
$u_2 = r i_1$
β
→
VCCS
δ
G_d
ε
λ
G_d
α
i_1
u
$r G_d^2 u$
u_2
γ
$i_1 = G_d u$
β

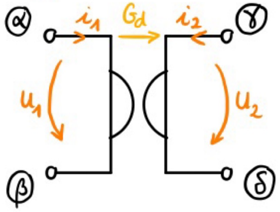
Stromgesteuerte Stromquelle


δ
α
i_1
i_2
γ
$i_2 = \beta i_1$
β
→
δ
G_d
ε
α
i_1
u
i_2
$\beta G_d u$
γ
$i_1 = G_d u$
β

Idealer Übertrager


α
ü : 1
δ
β
γ
→
α
G_d
ε
$\ddot{u} G_d$
δ
β
γ

Gyrator:

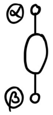

α
i_1
G_d
i_2
γ
u_1
u_2
β
δ

$i_1 = G_d (u_{k\gamma} - u_{k\delta})$

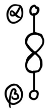
$i_2 = -G_d (u_{k\alpha} - u_{k\beta})$

$$\begin{array}{c|cccc} & α & β & γ & δ \\ \hline α & & & G_d & -G_d \\ β & & & -G_d & G_d \\ γ & -G_d & G_d & & \\ δ & G_d & -G_d & & \end{array}$$

- Einbau von Nullator und Norator



Spalten
u_k



Zeilen
i_q (有值的X)

Reaktive Netzwerkelemente

- **Ladung $q(t)$ $[C = As]$**

$$i(t) = \dot{q}(t) = \frac{d}{dt} q(t)$$

$$q(t) = q(t_0) + \int_{t_0}^{t} i(\tau)\, d\tau$$

$$= \int_{-\infty}^{t_0} i(\tau)\, d\tau + \int_{t_0}^{t} i(\tau)\, d\tau$$

$$= \int_{-\infty}^{t} i(\tau)\, d\tau$$

- **Fluss $\Phi(t)$ $[Wb = Vs]$**

$$u(t) = \dot{\Phi}(t) = \frac{d}{dt} \Phi(t)$$

$$\Phi(t) = \Phi(t_0) + \int_{t_0}^{t} u(\tau)\, d\tau$$

$$= \int_{-\infty}^{t} u(\tau)\, d\tau$$

- **resistiv, induktiv, kapazitiv, memristiv**

→ Nullator, Norator, Übertrager, NIK, LL, KS

- **resistiv, kapazitiv**

→ Spannungsquellen

- **resistiv, induktiv**

→ Stromquellen

- **memristiv, resistiv**. Gyrator

- **Beschreibung**

	Kapazität C	Induktivität L
implizit:	$f_c(u,q)=0$	$f_L(i,\Phi)=0$
explizit:	$q = c(u) = \chi(q)$	$\Phi = \ell(i) = \lambda(\Phi)$
linear:	$q = C\cdot u$	$\Phi = L\cdot i$
Dual:	$C \xrightarrow{R_d} L^d = C\cdot R_d^2$	$L \xrightarrow{R_d} C^d = \frac{L}{R_d^2}$
类比	G	R

- **Dualwandlung**

→ $L^d = C \cdot R_d^2$

→ $C^d = L \cdot \frac{1}{R_d^2}$

→ $\Phi^d = R_d \cdot q$

→ $q^d = \Phi \cdot \frac{1}{R_d}$

- **Eigenschaften reaktiver Eintore**

→ gedächtnisbehaftet: Ladung Fluss ✓

→ verlustlos: $W(t, t+T) = 0$

Kennlinie 无 Schleife

Kapazität: spannungs- oder ladungsgesteuert ⇒ verlustlos

Induktivität: strom- oder flussgesteuert ⇒ verlustlos

→ passiv: $P(t) = u(t)\cdot i(t) \geq 0$

Leistungsaufnahme 吸能

→ aktiv: $P(t) = u(t)\cdot i(t) < 0$

Leistungsabgabe 放能

- **Energie Arbeit $[J = VAs]$**

→ allgemein: $W = \int_{t_0}^{t_1} \underbrace{u(t)\cdot i(t)}_{p(t)}\, dt$

→ kapazitiv: (Kennlinie 和 q 轴 间的面积)

$$W_c(t_1, t_2) = \int_{t_1}^{t_2} u(t)\cdot i(t)\, dt$$

$$= \int_{q(t_1)}^{q(t_2)} \underbrace{u(q(t))}_{\text{用 q 表示 u}}\, dq$$

⇒ falls linear: $W_c = \frac{1}{2C}\cdot [q^2]_{q(t_1)}^{q(t_2)}$

→ induktiv: (Kennlinie 和 Φ 轴 间的面积)

$$W_c(t_1, t_2) = \int_{t_1}^{t_2} u(t)\cdot i(t)\, dt$$

$$= \int_{\Phi_1(t_1)}^{\Phi(t_2)} i(\Phi(t))\, d\Phi$$

⇒ falls linear: $W_L = \frac{1}{2L}\cdot [\Phi^2]_{\Phi(t_1)}^{\Phi(t_2)}$

- **gespeicherte Energie**

→ falls linear, im Punkt $(u_1, q_1) / (i_1, \Phi_1)$:

$$E_c(q_1) = E_c(u_1) = \frac{1}{2C}\cdot q_1^2 = \frac{C}{2}\cdot u_1^2$$

$$E_L(\Phi_1) = E_L(i_1) = \frac{1}{2L}\cdot \Phi_1^2 = \frac{L}{2}\, i_1^2$$

- **Kurzfragen**

→ $W > 0$. Energie aufnehmen, q ↗

→ $W < 0$. Energie abgeben, q ↘

Komplexe Wechselstromrechnung

- **Voraussetzungen**

→ sinusförmige Erregung

→ lineare, zeitvariante Elemente

→ im eingeschwungen Zustand

= Steady-State. 从固定f打

$\boxed{z.B}$ $A\cdot\cos(\omega t + \varphi_A) \rightarrow \boxed{LTI} \rightarrow B\cdot\cos(\omega t + \varphi_B)$

- **Lemma**

Lemma 1: Eineindeutigkeit

$\forall t: a(t) = b(t) \iff A = B$

$A_M \cos(\omega t + \alpha) = B_M \cos(\omega t + \beta) \iff A_M e^{j\alpha} = B_M \cdot e^{j\beta}$

Lemma 2: Linearität

$y(t) = \sum_{\ell=1}^{L} a_\ell x_\ell(t) \iff Y = \sum_{\ell=1}^{L} a_\ell X_\ell$ d

d.h. Linearkombination sinusförmiger Signale im Zeitbereich

$\shortparallel$

Linearkombination ihrer Zeiger im Frequenzbereich

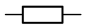
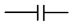
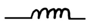
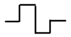
即 $y(t) = a(t) + b(t) \Rightarrow Y = A + B$

Lemma 3: Differentiation

$b(t) = \dot{a}(t) \iff B = j\omega \cdot A$

↳ Differentialoperator

- **Impedanz (Z) / Admittanz (Y)**

				
$Z = \frac{U}{I}$	$R = \frac{1}{G}$	$\frac{1}{j\omega C}$	$j\omega L$	M
$Y = \frac{I}{U}$	$G = \frac{1}{R}$	$j\omega C$	$\frac{1}{j\omega L}$	$\frac{1}{M}$

↳ $I = U j\omega C$

↳ $U = I j\omega L$

$\underline{I_c} = j\omega C \cdot \underline{U_c}$

$\underline{U_L} = j\omega L \cdot \underline{I_L}$

Zeiger!

- $\boxed{\text{Signal im Zeitbereich}}$

$\boxed{\text{komplexer Zeiger}}$

$$a(t) = \operatorname{Re}\{A \cdot e^{j\omega t}\}$$

$$a(t) = \hat{A}\cdot \cos(\omega t + \varphi) \longleftarrow A = \hat{A}\cdot e^{j\varphi}$$

- **Amplitude**: $\hat{A} = \sqrt{A \cdot A^*} = \sqrt{\operatorname{Re}^2\{A\} + \operatorname{Im}^2\{A\}}$

- **欧拉公式**: $e^{j\varphi} = \cos\varphi + j\sin\varphi$

- **Phase 相位**:

① $\operatorname{Re}\{A\} > 0$: $\varphi = \arctan\left(\frac{\operatorname{Im}\{A\}}{\operatorname{Re}\{A\}}\right)$

② $\operatorname{Re}\{A\} < 0$, $\operatorname{Im}\{A\} > 0$: $\varphi = \arctan\left(\frac{\operatorname{Im}\{A\}}{\operatorname{Re}\{A\}}\right) + \pi$

③ $\operatorname{Re}\{A\} < 0$, $\operatorname{Im}\{A\} < 0$: $\varphi = \arctan\left(\frac{\operatorname{Im}\{A\}}{\operatorname{Re}\{A\}}\right) - \pi$

Energie und Leistung [VAs]

→ Energie: $E = \int_0^T p(t)\,dt$

$= \int_0^T u(t)\,i(t)\,dt$

$= T \cdot P_w$

→ Mittlere Leistung: $P_w = \frac{E}{T}$

平均功率

$= \frac{1}{T}\int_0^T p(t)\,dt$

$= \frac{U_m^2}{2R}$

$= \frac{I_m^2 R}{2}$

→ Effektivwerte: $U_{eff} = \frac{U_m}{\sqrt{2}}$, $I_{eff} = \frac{I_m}{\sqrt{2}}$

有效值

→ 有效值与平均功率的关系: $P_w = U_{eff} \cdot I_{eff}$

$= \frac{U_{eff}^2}{R}$

$= R\ I_{eff}^2$

komplexe Leistung

→ komplexe Leistung: $P = \frac{1}{2} U I^* = \frac{1}{2} U^* I$

[VA]

$= P_w + j P_B$

→ Wirkleistung: $P_w = Re\{P\}$

[W]

→ Blindleistung: $P_B = Im\{P\}$

[Var]

→ Scheinleistung: $S = |P| = \sqrt{P_w^2 + P_B^2}$

[VA]

→ Mittelwert: $\bar{U} = \frac{1}{T}\int_0^T u(t)\,dt$

$\bar{I} = \frac{1}{T}\int_0^T i(t)\,dt$

mathematische Formeln

→ $\cos^2 x = \frac{\cos 2x + 1}{2}$

→ $\sin^2 x = \frac{1 - \cos 2x}{2}$

→ $e^{j\frac{\pi}{2}} = j$

→ $e^{j\pi} = e^{-j\pi} = -1$

→ $e^{-j\frac{\pi}{2}} = e^{j\frac{3\pi}{2}} = -j$

→ $e^{jx} + e^{-jx} = 2\cos x$

→ $e^{jx} - e^{-jx} = 2j\sin x$

Kurzfrage

→ Wieso kann eine Schaltung mit der Induktivität L nicht mithilfe der komplexen Zeiger analysiert werden?

=> L ist nicht linear!

补充

→ Zeigen Sie, dass ein Ohm'scher Widerstand memsristiv ist?

=> Memristiv bedeutet, dass eine Beschreibung der Form

$f(\Phi, q) = 0$ existiert.

=> Ohm'scher Widerstand: $u = Ri$ => Integration über t ($\int_{-\infty}^{t} dt$)

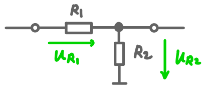
$\int_{-\infty}^{t} u\,dt = \int_{-\infty}^{t} Ri\,dt \Rightarrow \Phi = Rq$

→ VCVS mit Op-Amp:

① $\mu \geq 1$: Nicht invertierende Verstärker

② $\mu = 1$: Spannungsfolger

③ $0 < \mu < 1$: Spannungsfolger + Spannungsteiler + Spannungsfolger

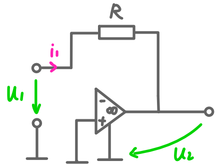


$$\boxed{U_2 = \frac{R_2}{R_1 + R_2} U_1}$$

④ $\mu \leq 0$: Spannungsfolger + invertierende Verstärker

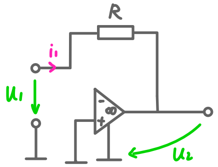
→ CCVS mit Op-Amp:

① $r = -R < 0$:



$$\boxed{U_2 = -RU_1}$$

② $r \geq 0$:



+ invertierende Verstärker

$$\boxed{U_2 = \frac{R_0 R}{R_1} U_1}$$

→ Quelle 在 Dualwandlung 时 {
闭合电路: 要换方向
不闭合电路: 不换方向